

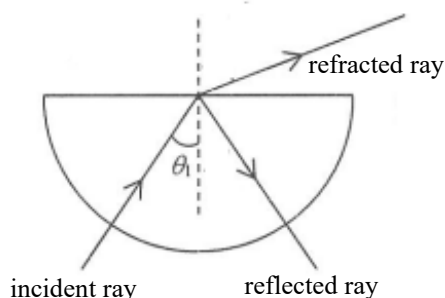
Total internal reflection

Two conditions for total internal reflection to occur:

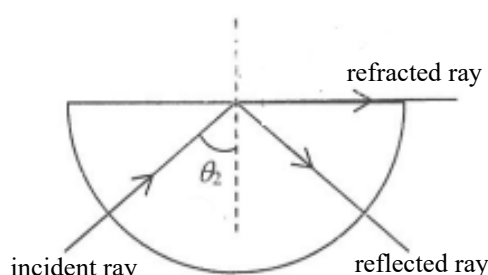
- Light travel from an **optically denser medium** to **less dense medium**.
- Angle of incidence i must be **greater than** the critical angle c .

$$\sin C = \frac{1}{n}$$

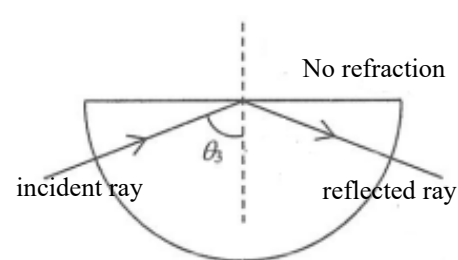
Angle of incidence (i) < critical angle (c)



(i) = (c)



(i) > (c)



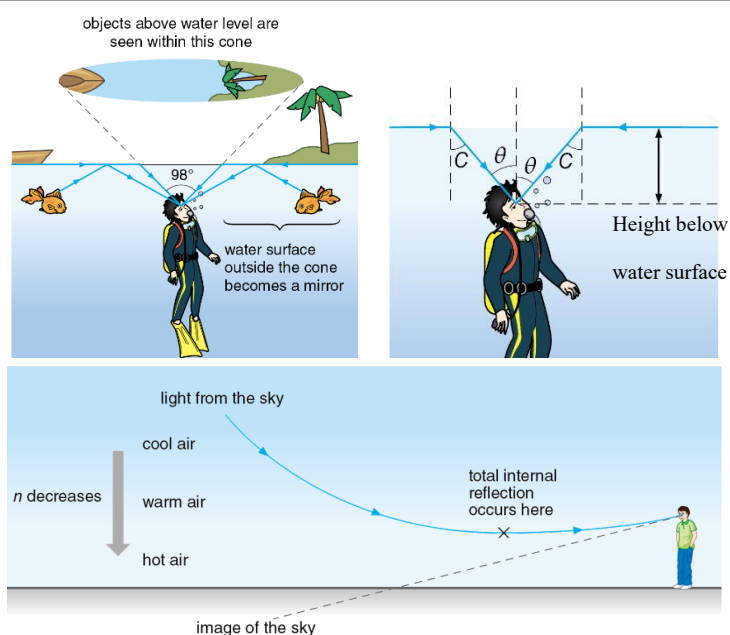
Daily example of total internal reflection

(1) Fish eye view

- A fish or a diver underwater can see objects above the water surface within a cone.
- Outside the cone, water surface looks a mirror.

(2) Mirage

- Road surface is hot and temperature gradually decrease up the road. Hot air has smaller refractive index.
- Light rays from distant object are gradually refracted away from normal as it approaches the ground.
- As angle of incidence is larger than critical angle, total reflection occurs.



Applications of total internal reflection

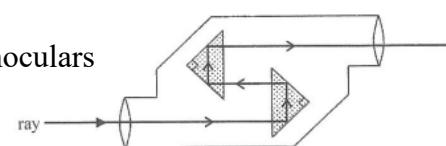
Advantages of using reflecting prism over plane mirror

- No multiple images
- The image will be brighter and clearer

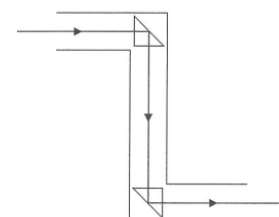
Advantages of using optical fibres over copper cables for telecommunications

- Optical fibres transmit signals with little loss of energy
- Optical fibres are lighter and thinner
- Optical fibres can carry more information

(i) Binoculars



(ii) Prismatic periscopes



(iii) Optical fibres



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1. Which of the following make(s) use of total internal reflection?

- (1) Mirage
- (2) Optical fibre
- (3) Mirror
- A. (2) only
- B. (1) and (2) only
- C. (2) and (3) only
- D. (1), (2) and (3)

B

2. Which of the following effects results from total internal reflection?

- A Swimming pools look shallower than they really are.
- B Diamonds sparkle when light shines on them.
- C You see your own image on the glass of a window when it is dark outside.
- D Reflection from a ruffled pond surface produces a distorted image of the surrounding environment.

B

3. Which of the following cases involve total internal reflection of light?

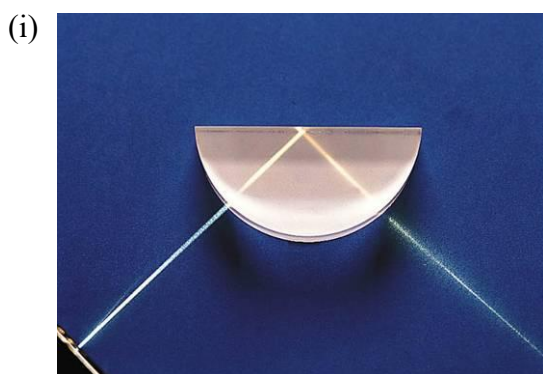


image of dolphin on water surface

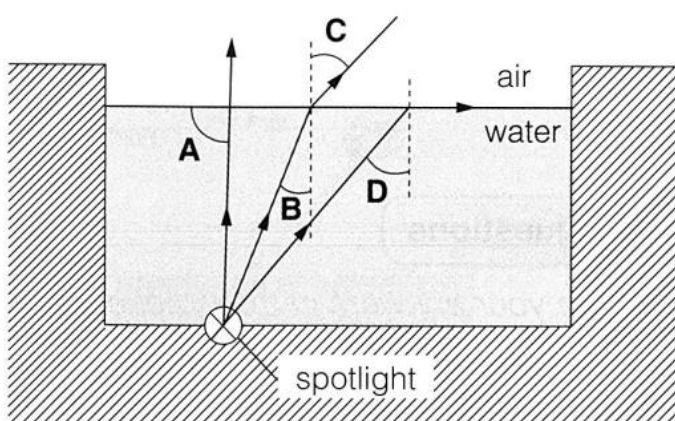


- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

A

image of building

4. A swimming pool is lit by a spotlight in the bottom of the pool. The directions of three rays from the spotlight are shown below. Which of the marked angles is the critical angle for the light?



D

6. The refractive index of a glass block is 1.5. What is the critical angle of the glass block?

- A. 0.667°
- B. 42°
- C. 48°
- D. 90°

B

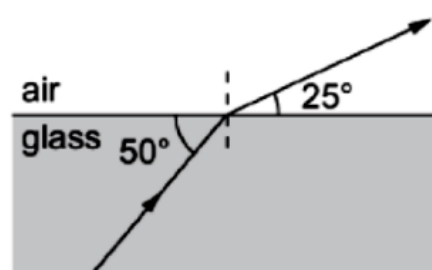
7. The refractive index of a glass block is 1.33. What is the critical angle of the glass block?

- A. 37°
- B. 42°
- C. 48.8°
- D. 70°

C

8. A ray of red light travels from a glass block to air as shown. Find the critical angle of the glass.

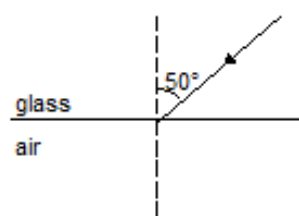
- A. 33.5°
- B. 41.1°
- C. 45.0°
- D. 45.2°



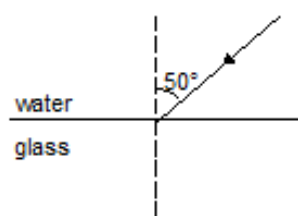
D

9. In which of the following cases will total internal reflection occur? Given that the refractive indices of glass and water are 1.5 and 1.33 respectively.

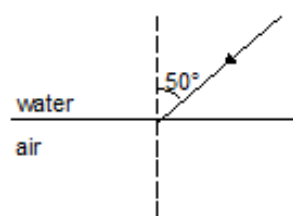
(1)



(2)



(3)



- A. (1) only
- B. (2) only
- C. (1) and (3) only
- D. (2) and (3) only

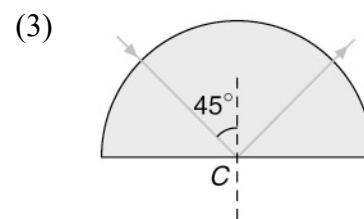
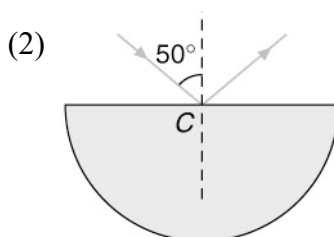
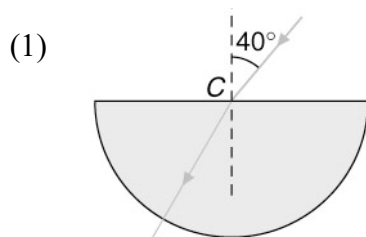
C

10. Medium A and B have refractive indices of 1.4 and 1.2 respectively. A light ray is now travelling in medium A. Total internal reflection will occur when the light ray is incident on a boundary formed by media A and B at angle of incidence larger than

- A. 31.0°
- B. 40.6°
- C. 49.4°
- D. 59.0°

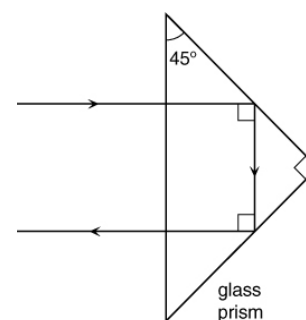
D

11. The following diagrams shows a light ray hitting a semicircular glass block in different ways.
C is the center of the semicircular block. Which diagram(s) is/are correct?
(Take critical angle of glass = 42° .)



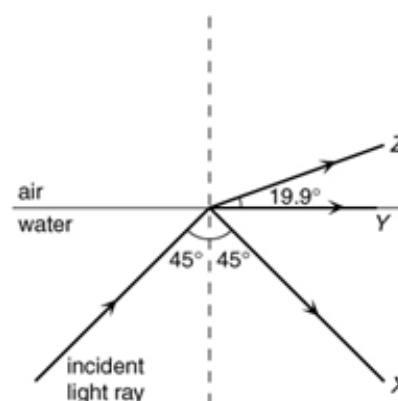
- A. (2) only
B. (1) and (2) only
C. (1) and (3) only
D. (1), (2) and (3)
12. The figure below shows the path of a light ray travelling through a prism. The refractive index of the glass must be larger than

- A. 1.00.
B. 1.33.
C. 1.41.
D. 1.52.

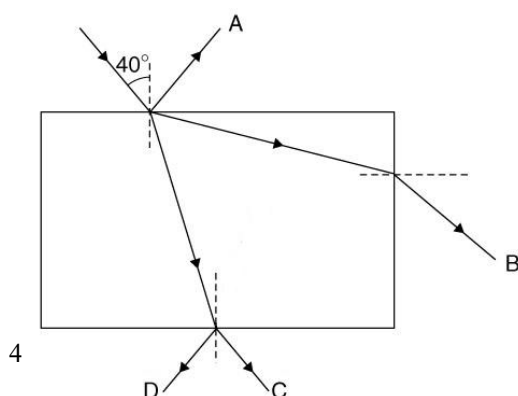


13. A ray of light strikes the water-air boundary at an angle of incidence of 45° as shown. It is known that the refractive index of water is 1.33. Which of the subsequent paths of the light ray is/are possible.

- A. X only
B. Z only
C. X and Y only
D. X and Z only



14. A light ray travelling in air hits the face of a rectangular glass block with the angle of incidence 40° . If the refractive index of glass is 1.5, which of the following best represents the path of the light ray?

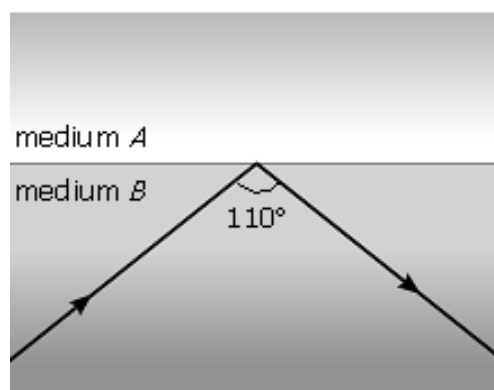


15. As shown in the figure below, a ray of light travelling in medium B strikes at the boundary between medium A and B and is totally reflected afterwards.

Which of the following statements **must** be correct?

- (1) Medium A is optically denser than medium B .
- (2) Light travels faster in medium A than in medium B .
- (3) The critical angle is 55° .

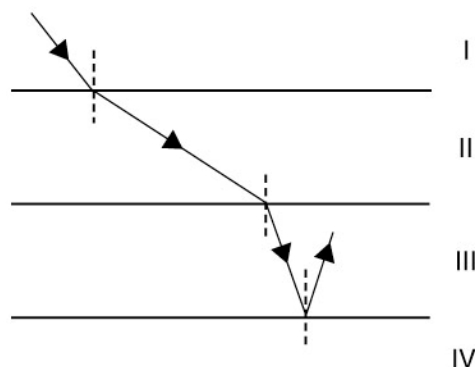
- A. (2) only
- B. (3) only
- C. (1) and (2) only
- D. (1) and (3) only



A

16. A light ray travels from medium I to medium IV, as shown in the following figure. Arrange the refractive indices of the media in ascending order.

- A. $IV < II < I < III$
- B. $III < IV < II < I$
- C. $IV < II < III < I$
- D. $III < I < IV < II$



A

17. Which of the following statements is/are correct?

(Given: refractive index of pure water = 1.33;
refractive index of peanut oil = 1.47)

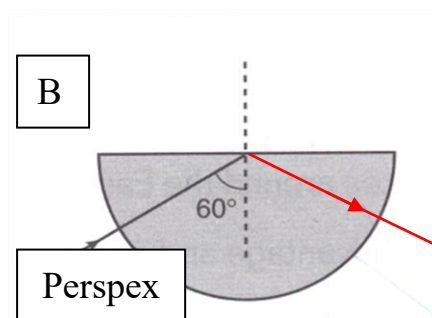
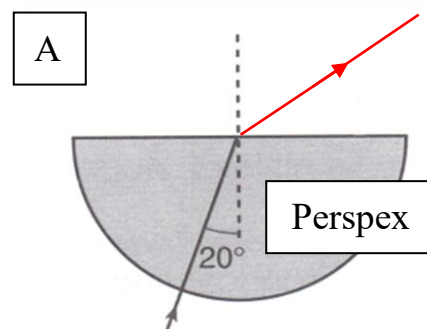
- (1) The apparent depth of a beaker filled with pure water is less than that of the same beaker filled with peanut oil.
- (2) The critical angle of peanut oil is smaller than that of pure water.
- (3) Total internal reflection may occur when light enters from peanut oil to pure water.

- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

D

18. The refractive index of the Perspex is 1.67.
- Find the critical angle of Perspex.
 - Sketch the emergent rays in Figure A and B.

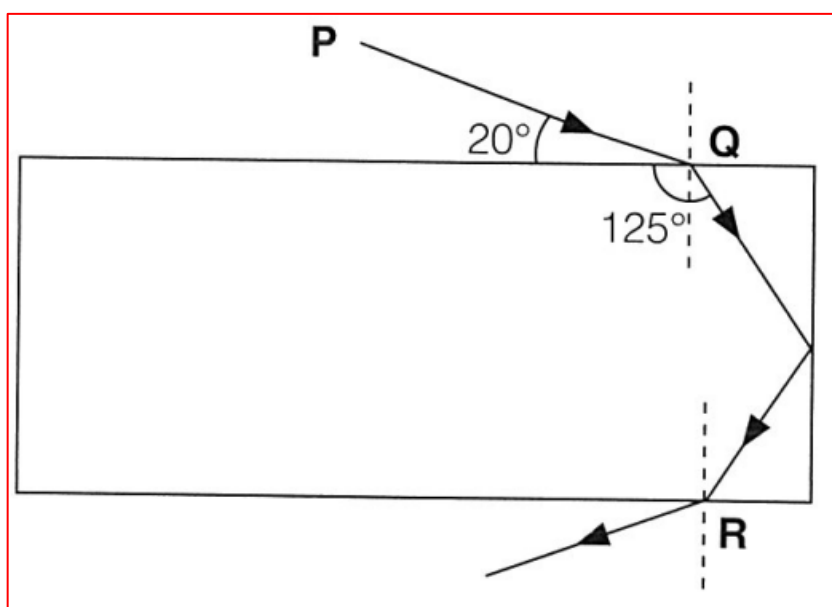
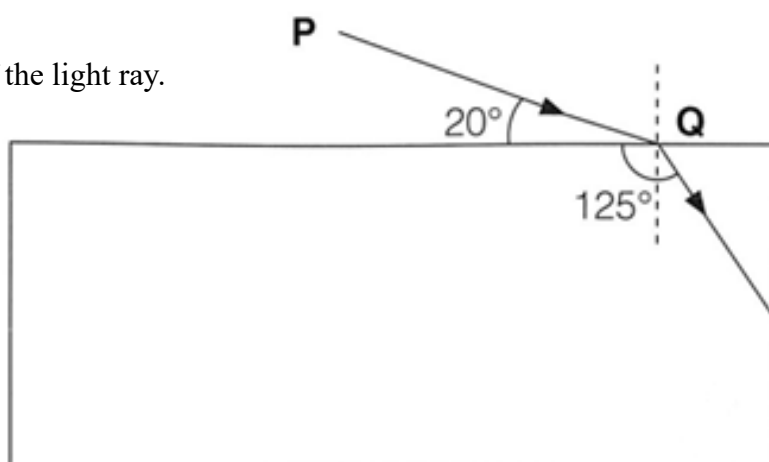
36.8°



19. A light ray PQ is incident on a glass block as shown in the figure below.

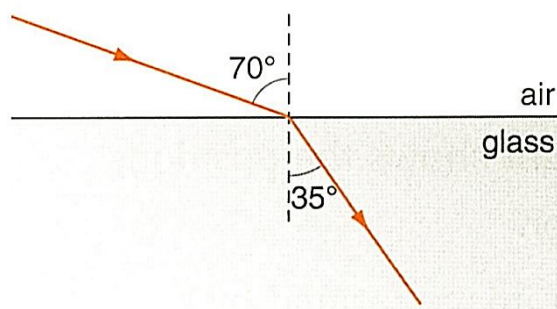
- Calculate the angle of incidence at point Q.
- Calculate the angle of refraction at point Q.
- Calculate the refractive index of glass.
- Calculate the critical angle of glass.
- Complete the diagram to show the path of the light ray.

- (a) 70°
(b) 35°
(c) 1.64
(d) 37.57°



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20. A light ray travels from air into glass as shown.

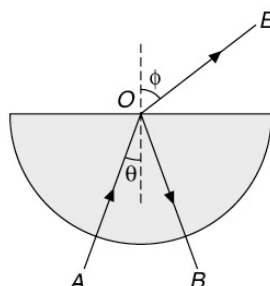


- Find the refractive index of the glass.
- If a light ray travels from the glass towards air and the angle of incidence is 50° , what will happen? Explain briefly.
- Then the glass is put in water. The refractive index of water is 1.33. Find the critical angle for the glass-water interface.

(a) Refractive index = $\frac{\sin \theta_a}{\sin \theta_g}$ $= \frac{\sin 70^\circ}{\sin 35^\circ}$ $= 1.638$ ≈ 1.64	1M 1A
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(b) Critical angle for the glass-air interface $= \sin^{-1}\left(\frac{1}{n}\right) = \sin^{-1}\left(\frac{1}{1.638}\right) = 37.6^\circ$ Since the angle of incidence is larger than the critical angle, total internal reflection takes place.	1A 1A
(c) By Snell's law, $n_g \sin \theta_g = n_w \sin \theta_w$ $1.638 \sin C = 1.33 \sin 90^\circ$ $C = 54.3^\circ$ $\therefore$ The critical angle is 54.3° .	1M 1A

21. Jack tries to find out the refractive index of a plastic. As shown in the figure below, he directs a light ray AO to enter the semicircular block made of the plastic. Part of the light is refracted into the air along OE and part of it is reflected along OB . After the experiment, he finds that the refractive index of the plastic is 1.58.



- Describe how the set-up could be used to measure the refractive index of the plastic.
- If the angle of incidence θ is 30° , what is the angle of refraction ϕ ?
- Calculate the critical angle of the plastic block.
- The angle θ is increased from 0° to 90° . Briefly describe the changes in the refracted ray OE .

- Measure several sets of readings of ϕ and θ .

Plot a graph of $\sin \phi$ against $\sin \theta$.

Fit the best straight line through the points.

The slope of the graph is the refractive index of the plastic.

- Apply Snell's law,

$$n = \frac{\sin \phi}{\sin \theta}$$

$$\sin \phi = n \sin \theta = 1.58 \sin 30^\circ$$

$$\phi = 52.2^\circ$$

- Critical angle of the plastic block

$$= \sin^{-1}\left(\frac{1}{n}\right)$$

$$= \sin^{-1}\left(\frac{1}{1.58}\right)$$

$$= 39.3^\circ$$

- As θ increases, the refracted ray OE turns clockwise.

When $\theta > 39.3^\circ$, OE disappears.

22. A ray of light travelling in air enters a semicircular plastic block with centre O as shown in Figure a. The angle of incidence at O is 35° . The refractive index of the plastic is 1.6.

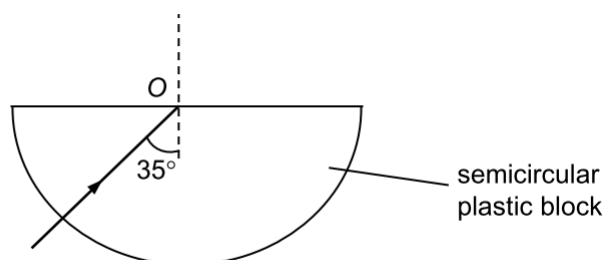
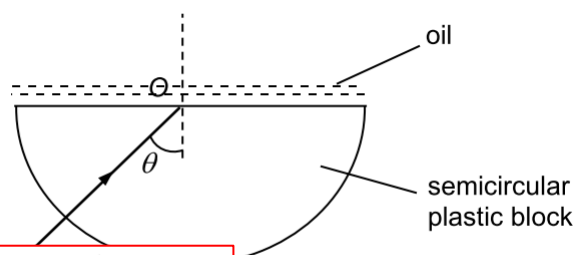
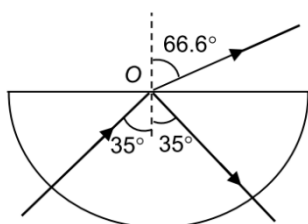


Fig a

- (a) When the ray reaches O , it splits into two.
- (i) Draw the two rays in Figure a. (2 marks)
- (ii) Mark the value of the angle between each ray and the normal in Figure a. (2 marks)
- (b) What happens if the ray reaches O with an angle of incidence greater than 40° ? Explain your answer. (5 marks)
- (c) A layer of oil (refractive index = 1.45) is applied on the flat side of the block (Fig b). If the light ray emerges from the plastic to the oil, what is the maximum value of the angle of incidence? (2 marks)



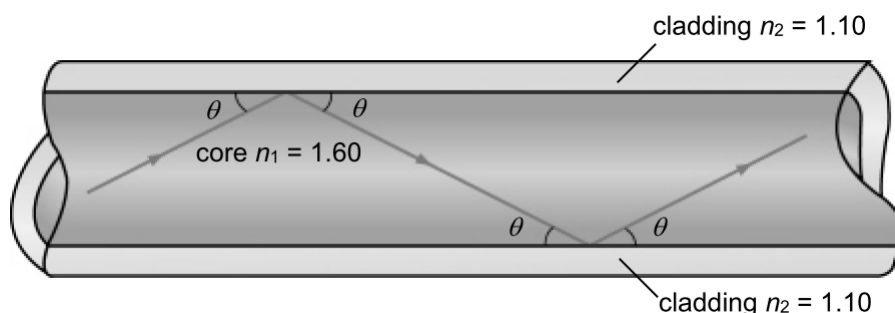
(a)



- (i) (Correct reflected ray) 1A
(Correct refracted ray) 1A
- (ii) (Correct angle of reflection) 1A
(Correct angle of refraction) 1A
- (b) Critical angle of the plastic-air interface $= \sin^{-1}\left(\frac{1}{n}\right)$ 1M
 $= \sin^{-1}\left(\frac{1}{1.6}\right) = 38.7^\circ$ 1A
- Since the light ray travels from plastic (optically denser) to air (optically less dense) 1A
and the angle of incidence is greater than the critical angle, 1A
the light ray undergoes total internal reflection. 1A
- (c) By Snell's law, 1M
 $n_1 \sin \theta_1 = n_2 \sin \theta_2$
 $1.45 \sin 90^\circ = 1.6 \sin C$
 $C = 65.0^\circ$ 1A
- The maximum angle of incidence is 65.0° .

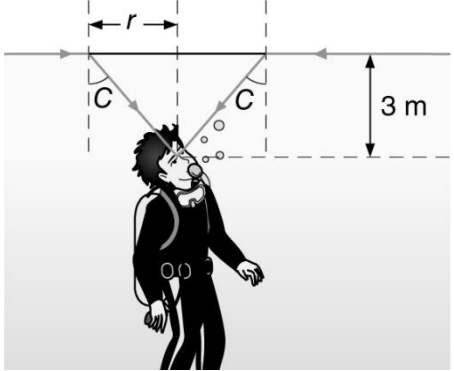
23. An optical fibre has an inner glass core surrounded by an outer glass cladding. A ray of light is totally internally reflected down the fibre as shown below, striking the walls at angle θ . The refractive indices of the core and the cladding are 1.60 and 1.10 respectively.

- (a) Give a reason why the glass core is surrounded by an outer glass cladding. (1 mark)
(b) Find the critical angles for the core and the cladding (both with respect to air). (3 marks)
(c) What is the maximum value of θ for which the ray will be retained within the fibre? (2 marks)



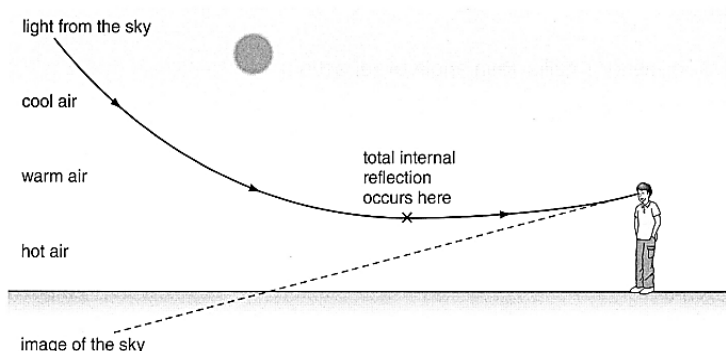
(a) Any one of the following: Protect the core from being damaged Reduce light leakage	1A
(b) Critical angle for the core = $\sin^{-1}\left(\frac{1}{n_1}\right)$ $= \sin^{-1}\left(\frac{1}{1.60}\right) = 38.7^\circ$	1M
Critical angle for the cladding = $\sin^{-1}\left(\frac{1}{1.10}\right) = 65.4^\circ$	1A
(c) By Snell's law, $n_1 \sin(90^\circ - \theta) = n_2 \sin 90^\circ$ $90^\circ - \theta = \sin^{-1}\left(\frac{1.10}{1.60}\right)$ $\theta = 46.6^\circ$ The maximum value of θ is 46.6° .	1M 1A

24. A diver is diving in the sea and his eyes are 3 m below the water surface. The refractive index of water is 1.33.
- Find the critical angle for the water-air interface.
 - As the diver looks upwards, the sky is compressed into a circular patch of radius r at the water surface. Find r .
 - As the diver dives deeper, will he find the radius of the circular patch in **(b)** larger or smaller? Explain briefly.

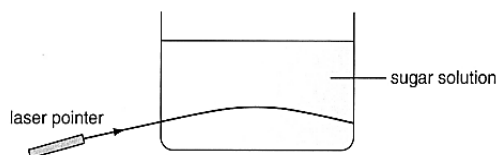
(a) Critical angle = $\sin^{-1} \frac{1}{1.33}$ $= 48.8^\circ$	1M
(b) With the notation in the figure below.	1A
	
$\tan C = \frac{r}{3}$	1M
$\tan 48.8^\circ = \frac{r}{3}$	1A
$r = 3.42 \text{ m}$	1A
(c) The diver will find the radius of the circular patch larger. The critical angle does not change with the depth of the diver in water. Therefore, the deeper the diver dives, the larger the radius of the circular patch is.	1A

25. On a hot sunny day, Chris finds that the ground at a distance ahead appears wet and reflective. In fact, the ground is dry and what he sees is the reflection of the sky. This type of illusion is called a mirage.
- (a) With the aid of a diagram, explain the formation of the mirage mentioned above.
- (b) With the aid of a diagram, describe an experiment to show a similar phenomenon in the laboratory.

- (a) On a hot sunny day, air near the ground is hotter and has a lower refractive index. 1A
With a continuous decrease of the refractive index, light from the sky is gradually refracted more towards the horizontal. 1A
When the light meets a layer of air at an angle beyond the critical angle, total internal reflection takes place. 1A
As a result, he sees the image of the sky when he looks down.



- (b) Put some sugar into a tank of water and let it dissolve naturally. 1A
The sugar solution at the bottom has a higher refractive index than the solution above. 1A
Direct a laser beam at a small angle above the horizontal into the solution at the bottom. The light ray will bend down due to refraction and total internal reflection. 1A



26. The refractive index of diamond is very large.

- (a) Explain briefly why a diamond sparkles.
- (b) Explain briefly why a diamond look less sparkling when it is put in water.

- (a) The critical angle of diamond is small and light undergoes total internal reflection in it frequently. With the facets carefully cut, most of the light entering the diamond is reflected at the bottom and emerges from the top. This makes it sparkle. •
- (b) The critical angle for a diamond-water interface is much larger than that for a diamond-air interface. Therefore, total internal reflection occurs less frequently when a diamond is put in water. Less light emerges from the top and the diamond becomes less sparkling.

MISCONCEPTION

Some students do not know that light must emerge from the top of the diamond to produce the sparkling effect. They wrongly think that light is reflected internally many times and trapped inside the diamond.